SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 1

Introduction to Operations & Supply Chain Management

Introduction to OSCM (Chapter 1). It opens with the Southwest Airlines case (why no assigned seats), then four learning objectives: (1) the elements of OSCM — design, operation, improvement; operations vs supply chain; the parka supply network; the five supply chain processes (plan, source, make, deliver, return); five differences between services and goods; the service package; the goods–services continuum and product–service bundling (IBM), (2) OSCM careers up to the COO and professional societies, (3) the history of major OSCM concepts, sustainability and the triple bottom line, business analytics and current issues, and (4) evaluating efficiency: efficiency vs effectiveness vs value, five Wall Street efficiency ratios, and Honda vs Toyota, GM and Ford.

Material

1 Learning objectives and where this chapter sits

1. **LO 1–1** — Identify the elements of OSCM.
2. **LO 1–2** — Know the career opportunities in OSCM.
3. **LO 1–3** — Recognise the major concepts that define the field.
4. **LO 1–4** — Evaluate the efficiency of a firm.

Part	Key question
I. Strategy, Products, and Capacity	How does strategy shape products and processes?
II. Manufacturing and Service Processes	How are transformation processes designed?
III. Supply Chain Processes	How are products moved through the supply chain?

IV. Supply and Demand Planning and Control	How are OSCM processes managed with computer systems?
V. Special Topics	How are the concepts used in special industries?

2 Opening case: efficiency at Southwest Airlines

How fast passengers board strongly affects an airline's cost. Southwest is the fastest at turning a plane around. It **does not assign seats**, aiming to keep planes in the air as much as possible despite many short flights per jet per day.

550
Jets

3,400
Flights per day

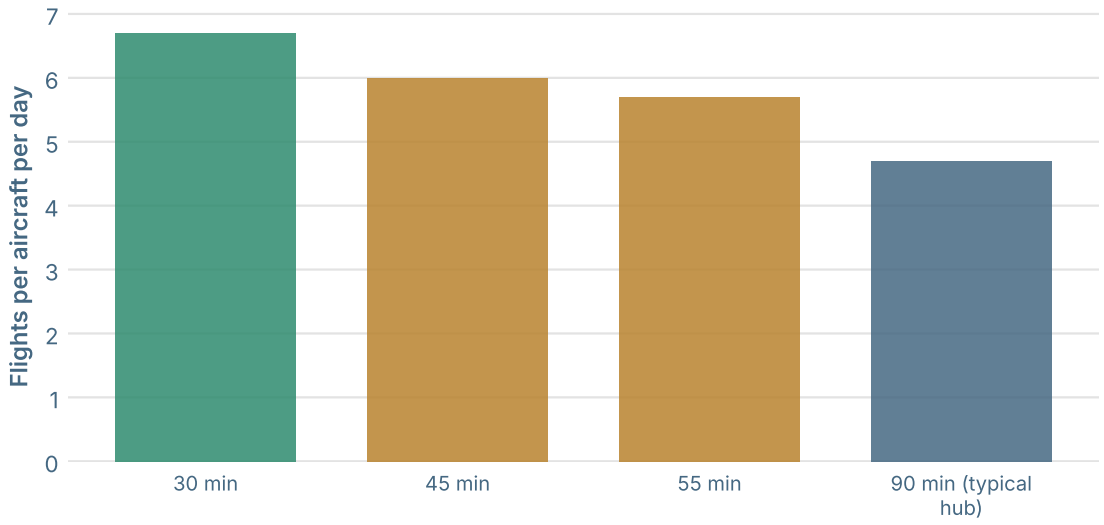
6+
Flights per jet per day

30–55 min
To turn a jet

≡ **Slide arithmetic**

At 45 minutes per turn, a jet still spends ~4.5 hours on the ground daily. Minutes saved loading passengers mean more flights without buying a single extra aircraft.

Turn time vs flights per aircraft per day



Illustrative: 16-hour day, 1.9 h average flight.

Discussion 1: (1) What does Southwest buy by not assigning seats, and who pays? *Faster turns; passengers pay with less certainty.* (2) Why not a 15-minute target? *Physical limits — baggage, cleaning, fuel, safety, crew; delays cascade.* (3) Would it work for a long-haul carrier flying one leg a day? *Little gain — the plane sits anyway, and long-haul passengers value assigned seats.* Name the process, the metric and the trade-off.

3 LO 1–1: what OSCM is — design, operation, improvement

OSCM is the **design, operation and improvement** of the systems that create and deliver a firm's primary products and services.

1. Design
Configure the network, facilities and process steps before anything is produced.

2. Operation
Plan and run capacity, labour and material so supply meets demand over time.

3. Improvement
Raise quality and remove waste continuously once the process runs.

A **process** is any step that transforms inputs — material, labour, information — into an output a customer values.

Exhibit 1.1 — one parka, four pipelines: blue (Polartec insulation: bought in bulk, finished, dyed/graded, stored), red (nylon extruded and drawn into yarn), green

(making the Supplex fabric), yellow (assembly, warehousing, pick & pack). Sketch the network end to end; each part is usually a different company, maybe in a different country, so coordination is the job.

	Operations	Supply chain
Definition	Manufacturing and service processes transforming resources into products customers want.	Processes moving information and material to and from those processes.
Includes	Manufacturing (cars, computers) and service (call centre, ER) processes; capacity, labour, material; quality and improvement.	Logistics, warehousing; input end (supply to plants) and output end (supply to customers).

Both sit inside a business strategy that matches customer preferences with supply-network realities, sustainably.

4 The five supply chain processes (Exhibit 1.3)

#	Process	Content	Hospital example
1	**Planning**	Processes to manage the resource base against forecast demand.	Forecasting patients, staffing plans
2	**Sourcing**	Select suppliers; receive and verify shipments; transfer; authorise payment.	Daily drug and supply deliveries
3	**Making**	Produce the product or provide the service; schedule workers; coordinate material.	Scheduling operations and blood tests
4	**Delivering**	Logistics: pick carriers, run warehouses, manage orders, invoice and collect.	ER staffed for on-demand service
5	**Returning**	Take back worn-out, defective or excess product; support customers after the sale.	Follow-ups, returning excess supplies

Discussion 2: Who owns a late parka's delivery date? (No single owner — coordination is the job.) Why hold inventory at four points? (Buffers between firms with different lead times and variability.) Map the five processes onto an ER.

5 Services vs goods: five differences and the service package

#	Difference	Explanation
1	**Intangible**	Can't be weighed; service innovations can't be patented, so a concept must expand fast before it's copied.
2	**Requires interaction**	However brief; face-to-face service needs facilities designed for the customer's presence.
3	**Heterogeneous**	Varies by hour and day with customers' and servers' attitudes; goods can have near-zero variability.
4	**Perishable & time dependent**	Can't be stored — you can't "come back last week" for a flight.
5	**A package of features**	Specified and evaluated as a package affecting the five senses.

Consequence

A defective good can be reworked or scrapped; a poor service experience has already been delivered.

Supporting facility
Location, decoration, layout, architecture, equipment.

Facilitating goods
Variety, consistency, quantity of physical goods (e.g. food with a meal).

Explicit services
Staff training, consistency, availability and access, comprehensiveness.

Implicit services
Attitude, atmosphere, waiting time, status, privacy and security, convenience.

6 The goods–services continuum and product–service bundling

Pure goods — food, chemicals, mining

→ Core goods — appliances, cars, data storage

→ Core services — hotels, airlines, ISPs

→ Pure services — university, medical, investment

Pure goods have become low-margin commodities, so they add services; core goods already include services (spare parts, dealer repair); core services integrate tangible goods (cable box); pure services still use facilitating goods (textbooks, spreadsheets).

Product–service bundling: building services into product offerings — maintenance, spare parts, training, even total systems design and R&D. **IBM** treats itself as a service business selling "business solutions". Successful implementers (1) consolidate services under one roof, (2) evolve from enhancing product performance to developing systems and modifications, (3) use them to move up the value stream into new markets.

! **Not for everyone**
Bundlers earn higher revenues but tend to lower profits as a percent of revenue than focused firms.

Discussion 3: sell the tractor or the harvested field (uptime)? When is lower % margin worth it (commoditised product, large installed base; judge by absolute profit and retention)? Which service trait hurts bundles most (often heterogeneity)?

7 LO 1–2: OSCM careers, the COO and professional societies

OSCM careers manage the planning, production and distribution of goods and services — on the buying side, hands-on and people-oriented.

Job	Main task
Plant manager	Workforce and physical resources for production
Hospital administrator	HR, staffing, supplies, finances
Supply chain manager	Vendor contracts, material flow
Logistics manager	Movement of goods
Warehouse/distribution manager	Replenishment, order fulfilment, staffing
Quality control manager	Acceptance sampling, control charts
Process improvement analyst	Lean tools to cut cycle time and waste
Project manager	New products, technology, new facilities
Call centre manager	Staffing and customer service

COO

- Sets competitive strategy with the CEO and president.
- Decides location, facilities, vendors, hiring policy.
- Supply chain, service and support are the hardest parts.

Professional societies

- APICS, CSCMP, ISM, PMI

8 LO 1–3: major OSCM concepts over time (Exhibit 1.5)

Period	Concept
Late 1970s	Manufacturing strategy
Early 1980s	Just-in-time (Japan)
Mid 1980s	TQM and quality certification
Early 1990s	Business process reengineering; service quality
Mid 1990s	Six Sigma
Late 1990s	Supply chain management and productivity
Early 2000s	E-commerce; service science
Mid 2010s	Business analytics

The 1980s revolution

JIT: high volume with minimal inventories arriving exactly when needed; combined with TQC (eliminating causes of defects) it is now called **lean manufacturing**. Ford's 1913 Model-T line came first.

9 Sustainability, the triple bottom line, business analytics and current issues

Sustainability: meeting current needs without compromising future generations. Three viabilities: **economic** (profit funds everything), **social** (job security, working conditions, development), **environmental** (non-polluting, non-depleting products and processes). The **triple bottom line** reports all three together.

Business analytics: using current business data to solve problems with mathematical analysis. What's new is the volume of data and accessible tools — statistics, forecasting, optimisation, possibly real time. Airline pricing example: real-time and historical demand plus models set prices by class, adjusted as departure nears (weather, fuel, crew), driving capacity utilisation. Analytics sets the seat price; operations sets how many times it can be sold.

Current issues: (1) coordinating separate but mutually supportive organisations amid surging outsourcing; (2) optimising global supplier, production and distribution networks using ERP data.

10 LO 1–4: efficiency, effectiveness, value and productivity

Efficiency
Doing something at the lowest possible cost.

vs

Effectiveness
Doing the right things to create the most customer value.

Value = a product's attractiveness relative to its cost; **labour productivity** and **asset productivity** measure output per unit of labour or assets.

↕ Top: effective · bottom: ineffective

Both
Southwest, Toyota.

Effective, inefficient
Loved but wasteful premium service; cash-burning startup.

Efficient, ineffective
The cheapest DVD factory in the streaming era.

Neither
Rarely survives.

↔ Left: efficient · right: inefficient

Company examples are illustrations, not from the slides.

11 Wall Street efficiency ratios and the Honda example

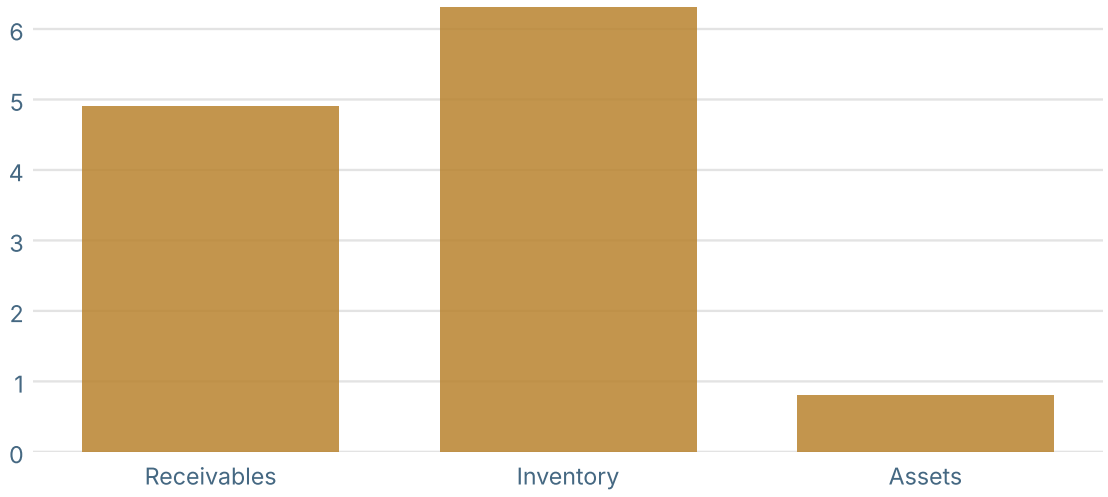
Ratios come from annual statements and vary greatly by industry — compare only within an industry.

Ratio	Formula	Meaning
Income per employee	`Net income ÷ headcount`	Profit per employee

Revenue per employee	`Sales ÷ headcount`	Sensitive to outsourcing
Receivables turnover	`Sales ÷ receivables`	Speed of collection
Inventory turnover	`COGS ÷ average inventory`	Times inventory is sold and replaced
Asset turnover	`Sales ÷ total assets`	Most comprehensive

> **Start with asset turnover**
> It includes plants, warehouses and equipment, so it is strongly affected by technology investment and outsourcing.

Honda turnover ratios (x)



Example 1.1 — Honda: income per employee US$30,203; revenue per employee US$585,647; receivables 4.9x; inventory 6.3x; asset 0.8x. GM has the highest asset turnover (recently bankrupt and recapitalised); Honda is fine vs the industry, stronger than Ford, weaker than Toyota and GM; Ford leads inventory turns but trails on receivables.

Discussion 4: best ratio ≠ best-run (GM's shrunken assets); Ford's slow receivables may reflect vehicle financing; outsourcing inflates revenue per employee and asset turnover without real improvement; pair any ratio with an operational metric for the COO.

12 Wrap-up: five things to carry out of Chapter 1

1. **OSCM is design, operation and improvement.** Capacity and cost are created by process choices long before any report.
2. **Sketch the network before you optimise it.**
3. **Services are not goods with a smile.**
4. **Every concept arrived to solve a problem.**
5. **Measure efficiency, then interrogate the measure** — within industry, over time, beside an operational metric.

Key takeaways

1. OSCM = design, operation and improvement of systems that create and deliver primary products/services; a process transforms inputs into valued outputs.
2. Operations transform resources; the supply chain moves information and material to and from operations.
3. Sketch the supply network end to end (parka example) before optimising.
4. Five supply chain processes: plan, source, make, deliver, return — also in services.
5. Five service differences: intangible, interaction, heterogeneous, perishable, package of features.
6. Goods–services continuum; bundling (IBM) raises revenue but lowers % margin.
7. Careers up to COO; societies: APICS, CSCMP, ISM, PMI.
8. Concepts over time: manufacturing strategy, JIT, TQM, BPR, Six Sigma, SCM, e-commerce, analytics; JIT + TQC = lean.
9. Sustainability and the triple bottom line; business analytics.
10. Efficiency vs effectiveness vs value.
11. Five ratios; asset turnover first; compare within industry and pair with an operational metric.

Case studies

Case & worked answers

Southwest Airlines — why not assign seats?

550 jets, 3,400 flights/day, 30–55 minute turns. Unassigned seating speeds boarding, raising flights per aircraft (asset utilisation); passengers pay with less certainty. Below some point physical and safety limits bite. Long-haul carriers fly

one leg a day and sell comfort, so the design does not transfer. Process = turnaround; metric = ground minutes; trade-off = comfort vs utilisation.

Case & worked answers

Discussion: the parka, the hospital and IBM

No single firm owns the parka's delivery date — coordination is the job; buffers decouple stages. Selling "uptime" fits when the product commoditises and the firm has service capability; measure retention and service margin. Heterogeneity and perishability make bundles hardest.

Practice questions

Try answering first. The answer key is below each question.

1. OSCM consists of...

A. Plan, do, check
B. Design, operation, improvement
C. Buy, make, sell
D. Source, make, deliver

Answer: B. As defined.

2. The process handling defective or excess goods from customers is...

A. Plan
B. Source
C. Deliver
D. Return

Answer: D. Return.

3. Why can't services be stored?

A. Intangible
B. Perishable — unused capacity is lost
C. Heterogeneous
D. Needs interaction

Answer: B. Empty seats vanish.

4. The most comprehensive efficiency ratio is…

A. Income/employee
B. Receivables turnover
C. Inventory turnover
D. Asset turnover

Answer: D. Includes all assets.

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5. Faster turns without new planes most improve…

A. Debt ratio
B. Asset turnover
C. Current ratio
D. Payout

Answer: B. More output, same assets.

Sources

- P3MD MK4 OSCM — Introduction to Operations and Supply Chain Management (class slides)
- Jacobs & Chase — Operations and Supply Chain Management, Ch. 1 (Exhibits 1.1–1.5, Example 1.1)